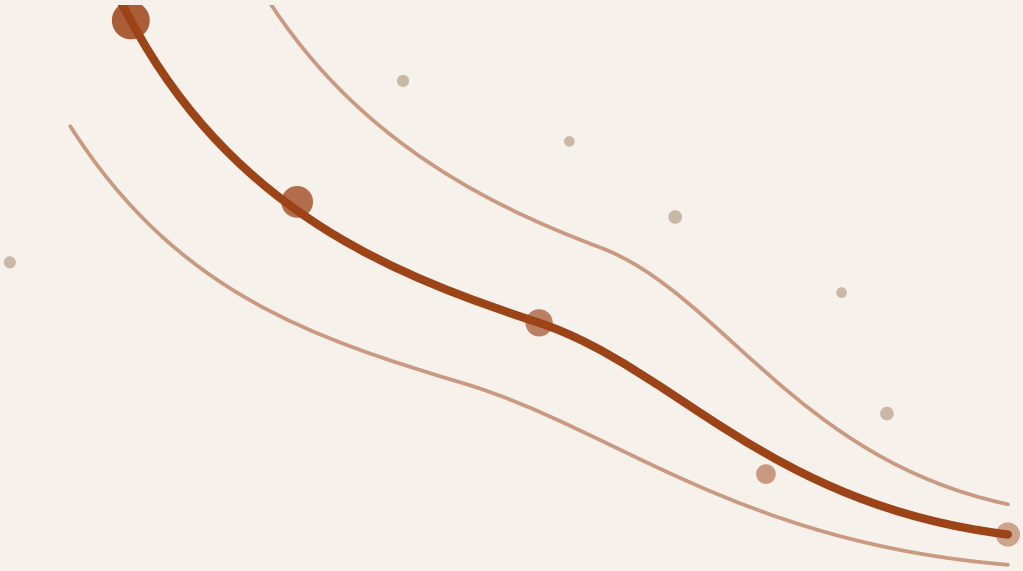
Innovation Management

From scattered initiatives
to a sustained organizational capability

Dr. Omar A. Alsamani



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Back matter

Glossary

The terms defined in the chapters of the book, with the chapter in which each definition appears. Entries are listed alphabetically.

Absorptive capacity Absorptive capacity	An organization's ability to recognize the value of external knowledge, assimilate it and apply it. Chapter 18
Academic engagement Academic engagement	Faculty collaboration with external organizations through joint or contract research or consulting, as distinct from the commercialization of research. Chapter 20
Academic heartland Academic heartland	The departments and colleges where teaching and research take place, as distinct from peripheral units. Chapter 20
Architectural innovation Architectural innovation	Linking known components in a new way. Chapter 1
Boundary-spanning activities Boundary-spanning activities	A team's activities with external parties: communicating with leadership, coordinating with other units and gathering knowledge. Chapter 9
Business model Business model	The logic by which an organization delivers value to those it serves and captures enough to sustain

	itself. Chapter 13
Business process innovation Business process innovation	A new or improved change in the way an organization operates in one of its business functions. Chapter 1
Campbell's law Campbell's law	The more a quantitative indicator is relied on in decision-making, the greater the pressure to manipulate it and to distort what it measures. Chapter 16
Certification Certification	Verification by an independent body that an organization's system conforms to the requirements of a specified standard. Chapter 5
Challenge question Challenge question	A formulation of a priority problem that specifies what is needed without imposing the solution. Chapter 4
Chief innovation officer Chief innovation officer	The person who leads the orchestrating unit and achieves results through units they do not head. Chapter 6
Collingridge dilemma Collingridge dilemma	The difficulty of controlling an innovation: it is easy to change while its effects are unknown, and hard to change once they become clear. Chapter 19
Componential model of creativity Componential model of creativity	A model that explains individual creativity through domain-relevant skills, creativity-relevant processes and intrinsic task motivation, within the context of the work environment. Chapter 7
Concept Concept	A description of a solution clear enough to be understood and tested, including its critical

	assumptions. Chapter 12
Convergent thinking Convergent thinking	Arriving at the most suitable answer among the alternatives. Chapter 12
Creative climate Creative climate	The patterns of behavior and feeling observed in daily work that encourage or hinder creativity. Chapters 8 and 16
Creative self-efficacy Creative self-efficacy	A person's belief in their ability to produce new and useful ideas. Chapter 7
Critical assumption Critical assumption	An assumption that, if wrong, brings down the whole initiative. Chapter 14
Culture-embedding mechanisms Culture-embedding mechanisms	The practices through which people learn the actual culture, such as measurement, rewards and responses to crises. Chapter 8
Decision limbo Decision limbo	The state of an initiative that is neither accepted nor rejected, and so remains suspended without learning or impact. Chapter 5
Decision rights Decision rights	Specifying who holds the authority to make each type of innovation decision. Chapter 10
Degree of novelty Degree of novelty	The novelty of a solution relative to a unit of reference: the organization, the market or the world. Chapter 1
Design fixation Design fixation	The pull of thinking toward the features of a known solution or the first solution proposed. Chapter 12

Disciplined collaboration Disciplined collaboration	Collaboration between units directed at a result whose value exceeds its cost. Chapter 6
Discovery-driven planning Discovery-driven planning	Planning that states assumptions explicitly and sequences the stages so that the most important are tested first. Chapter 14
Disruptive innovation Disruptive innovation	Innovation that starts at the low end of a market or in a new market, then improves until it competes with prevailing solutions. Chapter 1
Divergent thinking Divergent thinking	Producing multiple, varied answers to a single question. Chapter 12
Dominant logic Dominant logic	The way of thinking that the existing model imposes on the evaluation of new opportunities. Chapter 13
Dynamic capabilities Dynamic capabilities	An organization's ability to sense and seize opportunities and to reconfigure its resources and practices in a changing environment. Chapter 2
Ecosystem risk Ecosystem risk	The dependence of an innovation's success on partners and links in the adoption chain that the organization does not control. Chapter 18
Employee voice Employee voice	Employees' expression of ideas, suggestions and concerns with the aim of improving work. Chapter 3
Escalation of commitment Escalation of commitment	Continuing to invest in a failing course of action to justify the original decision. Chapter 10

Executive sponsor Executive sponsor	A leader who links the initiative to strategy, gives it legitimacy and resources, and removes obstacles in its way. Chapter 3
Explainability Explainability	The ability of those using an automated system to explain the reasons behind its decisions. Chapter 19
Exploitation Exploitation	Refining and deepening what the organization already knows in order to achieve efficiency. Chapter 3
Exploration Exploration	Searching for new knowledge and opportunities under uncertainty. Chapter 3
External realities External realities	The conditions surrounding the organization and how they change, in markets, society, legislation and technology. Chapter 2
Front end of innovation Front end of innovation	The phase before a project is formally launched, including identifying needs, exploring opportunities and generating ideas. Chapter 1
Front-loading Front-loading	Identifying and solving problems in the early stages, when change is cheaper. Chapter 17
Hard core and soft periphery Hard core and soft periphery	The distinction between what must remain fixed in an innovation and what can be adapted to each context. Chapter 15
Heavyweight team Heavyweight team	A team with a leader who holds strong authority and dedicated members, suited to innovation that requires integration across units. Chapter 9

Implementation climate Implementation climate	People's perception that using the innovation is expected, supported and rewarded. Chapter 15
Incremental innovation Incremental innovation	Innovation that builds on an existing offering or capability without changing its fundamental nature. Chapter 1
Influence without authority Influence without authority	A person's ability to move people who do not report to them, through reciprocity and partnership building. Chapter 6
Innovation Innovation	A new or significantly improved application that is adopted by users and creates value. Chapter 1
Innovation ambassador network Innovation ambassador network	People in different units who help colleagues articulate their ideas and connect the front line with the innovation system. Chapter 9
Innovation campaign Innovation campaign	A time-bound call for ideas on a specific challenge question, with a published mechanism for responding. Chapter 9
Innovation capability Innovation capability	An organization's ability to innovate repeatedly without depending on particular individuals. Chapter 6
Innovation implementation Innovation implementation	The consistent, actual use of an innovation by its intended users after the decision to adopt it. Chapter 15
Innovation inputs / outputs Innovation inputs / outputs	Inputs are the resources and conditions that enable innovation; outputs are its knowledge, technological and creative results. Chapter 2

Innovation lab Innovation lab	A unit dedicated to experimentation and to tackling complex problems with new methods. Chapter 6
Innovation management Innovation management	The practices and capabilities that steer the flow of opportunities, ideas and initiatives, and manage experimentation, decisions and resources through to the realization of value. Chapter 1
Innovation management system Innovation management system	The formal framework through which an organization manages the flow of opportunities and initiatives through to value, encompassing strategy, governance, roles, resources and measurement. Chapter 5
Innovation process Innovation process	The successive stages an initiative passes through, from opportunity to implemented solution. Chapter 5
Innovation strategy Innovation strategy	A set of interlinked choices that define why an organization innovates, where, with what ambition and by what means. Chapter 4
Innovation theater Innovation theater	Activities that resemble innovation in appearance without bringing about change in products, services or ways of working. Chapter 1
Institutionalization Institutionalization	Embedding an initiative in the organization's routine systems so that it endures without relying on the enthusiasm of individuals. Chapter 20
Intelligent failure Intelligent failure	An unexpected result of a small, well-designed experiment in uncharted territory that yields new knowledge. Chapter 8

Internal realities Internal realities	The organization's current state in terms of leadership, culture, structures, roles and skills. Chapter 2
Intrapreneurship Intrapreneurship	Entrepreneurial behavior by individuals within an existing organization to lead a new innovation. Chapter 1
Intrinsic task motivation Intrinsic task motivation	Being drawn to work by its interest and challenge rather than by an external reward. Chapter 7
Isomorphic mimicry Isomorphic mimicry	Adopting the forms of successful organizations without the capabilities that make them work. Chapter 20
Issue selling Issue selling	Employees' efforts to draw senior management's attention to issues they consider important and to shape how management understands them. Chapter 6
Job-relevant diversity Job-relevant diversity	Differences among team members in disciplines and expertise relevant to the problem. Chapter 9
Jobs to be done Jobs to be done	The progress a person seeks in a particular circumstance, enlisting a product or service to achieve it. Chapter 11
Lead users Lead users	People who face a need before others and expect large benefits from meeting it, and so innovate solutions for themselves. Chapter 11
Loose coupling Loose coupling	The condition of units that respond to one another while each retains its independence, so that what

	happens in one carries over only weakly to the others. Chapter 20
Maturity model Maturity model	A description of successive levels of capability, used to assess the current position and identify the next step. Chapter 16
Micro-credential Micro-credential	Formal recognition of specific competencies acquired in a short learning program. Chapter 20
Mimetic isomorphism Mimetic isomorphism	The tendency of organizations, under ambiguity, to imitate organizations they see as successful or legitimate. Chapter 20
Minimum viable product Minimum viable product	The smallest version of a product that allows the greatest amount of learning with the least effort. Chapter 14
Mission-oriented innovation Mission-oriented innovation	Mobilizing efforts and resources toward a specific, ambitious outcome. Chapter 4
National innovation system National innovation system	The network of public and private institutions whose interactions produce, diffuse and use knowledge in a country. Chapter 18
Not-invented-here syndrome Not-invented-here syndrome	Resistance to ideas and knowledge that come from outside the team or the organization. Chapter 18
Open innovation Open innovation	A distributed innovation process based on knowledge flows purposively managed across organizational boundaries. Chapter 18

Optimism bias Optimism bias	The tendency, when planning, to underestimate costs and risks and to overstate benefits. Chapter 17
Orchestrating unit Orchestrating unit	A unit linked to top leadership that connects the components of the existing system without owning their initiatives. Chapter 6
Organizational ambidexterity Organizational ambidexterity	An organization's ability to combine exploitation and exploration at the same time. Chapter 3
Organizational culture Organizational culture	The shared assumptions that a group has learned and passes on to new members as the correct way to think and act. Chapter 8
Perceived organizational support Perceived organizational support	An employee's belief that the organization values their contribution and cares about their well-being. Chapter 3
Pilot Pilot	A limited run of the solution in a near-real setting to test its value and adoption before scaling. Chapters 5 and 14
Problem finding Problem finding	The process by which a person identifies and formulates a problem before solving it; it is considered part of creativity. Chapter 11
Problem-driven iterative adaptation Problem-driven iterative adaptation	An approach that builds solutions locally, starting from problems, through experimentation, learning and continuous adaptation. Chapter 20
Procedural justice Procedural justice	The fairness of the way a decision is made, grounded in involvement, explanation and clear

	expectations. Chapter 10
Production blocking Production blocking	The drop in idea generation in groups because members must wait their turn to speak. Chapter 12
Prototype Prototype	A representation of a solution built to answer a specific question about its role, its look and feel, or its implementation. Chapter 14
Proxy measure Proxy measure	A measure used in place of the intended construct because the latter is hard to measure directly; it may carry hidden biases. Chapter 19
Psychological safety Psychological safety	A shared belief within a team that asking questions, raising objections and admitting mistakes will not bring punishment or embarrassment. Chapter 8
Public value Public value	What society and its members value in the outcomes of public services and policies. Chapters 1 and 13
Randomized controlled trial Randomized controlled trial	An experiment in which participants are randomly assigned to the solution or an alternative, in order to isolate the solution's effect. Chapter 14
Real options Real options	A logic that treats a small early investment as the right to scale up later or to stop with a limited loss. Chapter 10
Reference class forecasting Reference class forecasting	Estimating cost and duration by looking at what actually happened in similar projects. Chapter 17

Reframing Reframing	Formulating a problem from a different angle that reveals new solutions. Chapter 11
Relevance Relevance	The continuing connection between what an organization offers and the needs of its beneficiaries, now and in the future. Chapter 2
Responsible innovation Responsible innovation	Taking care of the future through collective stewardship of science and innovation in the present, grounded in anticipation, reflexivity, inclusion and responsiveness. Chapter 19
Results chain / logic model Results chain / logic model	A sequence linking inputs and activities to outputs, outcomes and impact. Chapter 16
Scenario planning Scenario planning	Sketching several plausible futures in order to test decisions against them. Chapter 4
Search field / Innovation arena Search field / Innovation arena	A priority area of opportunity that directs exploration efforts. Chapter 4
Servant leadership Servant leadership	A leadership style that makes serving followers and meeting their needs and growth an end in itself. Chapter 3
Solution requirements Solution requirements	What any solution must achieve to count as successful, regardless of its form. Chapter 12
Spending efficiency Spending efficiency	Achieving the greatest possible value with the available resources. Chapter 17

Stage-Gate Stage-Gate	A system that divides an initiative's journey into stages separated by decision points with predefined criteria. Chapter 10
Staged funding Staged funding	Funding an initiative stage by stage, with each tranche tied to passing a gate. Chapter 10
Stages of concern Stages of concern	The progression of adopters' questions and concerns from the self, to the task, to impact. Chapter 15
Strategic triangle Strategic triangle	A framework that links the success of a public initiative to the alignment of public value, legitimacy and support, and operational capacity. Chapter 13
Student innovator journey Student innovator journey	An extended path that connects the university's units around what the student needs at each stage. Chapter 20
Sustainability-oriented innovation Sustainability-oriented innovation	Intentional changes to an organization's values, products and processes that create environmental and social value alongside economic value. Chapter 19
The innovator's path The innovator's path	A model that reads an idea's journey from its owner's perspective, from thought to practice, together with its outcomes and its surrounding environment. Chapter 7
Three horizons Three horizons	Dividing initiatives according to their distance from the current business and their degree of uncertainty. Chapter 4

Total cost of ownership Total cost of ownership	The cost of a solution over its whole life, from purchase through operation to retirement. Chapter 17
Triple helix Triple helix	A model that explains innovation through the interaction of university, industry and government. Chapter 18
Unintended consequences Unintended consequences	Consequences of purposive action that the actor neither intended nor foresaw. Chapter 19
Value proposition Value proposition	The job a solution helps people get done, and why it is better than their alternatives. Chapter 13
Vanity metrics Vanity metrics	Numbers that look good in reports but do not indicate the value actually achieved. Chapter 16
Voltage effect Voltage effect	The decline in a program's impact as it scales from a pilot to wide implementation. Chapter 15
VUCA VUCA	A description of the environment along four dimensions: volatility, uncertainty, complexity and ambiguity. Chapter 2
Weak signals Weak signals	Early, ambiguous information about an approaching change. Chapter 4
Wicked problem Wicked problem	A problem that cannot be definitively formulated and that changes with every attempt to solve it. Chapter 11
Work environment for creativity	The enablers and obstacles within an organization that affect employees' creativity, particularly their motivation. Chapter 7

Work environment for creativity	
Workaround Workaround	A method people devise to cope with a problem that no one has solved for them. Chapter 11

How does an organization move from scattered initiatives that celebrate innovation to a sustained capability to produce it?

This book traces the path of innovation from the first signal to a solution adopted in practice that creates value, and analyzes the organizational foundations that make this path a settled practice: strategy, systems, culture, governance, teams and measurement, with people at the center.

It rests on three pillars: a clear, high-priority problem; motivated people whose needs are met; and a system that connects them and turns effort into repeatable results. It combines international research evidence with the Saudi context, and pairs each chapter with a tool that can be used in a real organization.

Its twenty chapters are organized in five parts: foundations; institutional foundations and people; the initiative journey; value and the ecosystem; and an application to the university innovation ecosystem.

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